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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 3rd April Evening Shift

JEE Main Question Paper with Answer Key

3 Apr 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	25 questions
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Answer key	at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I : Wet cotton clothes made of cellulose based carbohydrate takes comparatively longer time to get dried than wet nylon polymer based clothes.

Statement II : Intermolecular hydrogen bonding with water molecule is more in nylon-based clothes than in the case of cotton clothes.

In the light of above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q2. MCQ (Single Correct Answer)

Open solution

Fat soluble vitamins are:

A. Vitamin B_1

B. Vitamin C

C. Vitamin E

D. Vitamin B 12

E. Vitamin K

Choose the correct answer from the options given below:

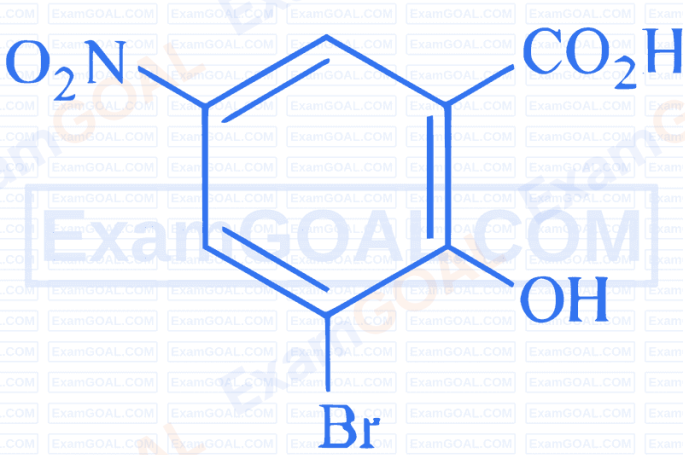
(A) B & C Only **(B)** C & D Only **(C)** C & E Only **(D)** A & B Only

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Q3. MCQ (Single Correct Answer)

Open solution

What is the correct IUPAC name of



?

(A) 3-Bromo-2-hydroxy-5-nitrobenzoic acid
(B) 2-Hydroxy-3-bromo-5-nitrobenzoic acid
(C) 3-Bromo-4-hydroxy-1-nitrobenzoic acid
(D) 5-Nitro-3-bromo-2-hydroxybenzoic acid

Q4. MCQ (Single Correct Answer)

Open solution

Mass of magnesium required to produce 220 mL of hydrogen gas at STP on reaction with excess of dil. HCl is

Given: Molar mass of Mg is $24\ \mathrm{g\ mol^{-1}}$.

(A) 0.24 mg
(B) 235.7 g
(C) 2.444 g
(D) 236 mg

Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I : When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II : At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is not increased at melting point.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false
(B) Statement I is true but Statement II is false
(C) Both Statement I and Statement II are true
(D) Statement I is false but Statement II is true

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Q6. MCQ (Single Correct Answer)

Open solution

The correct orders among the following are

A. Atomic radius : $\mathrm{B} < \mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

B. Electronegativity : $\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl} < \mathrm{B}$

C. Density : $\mathrm{Tl} < \mathrm{In} < \mathrm{Ga} < \mathrm{Al} < \mathrm{B}$

D. 1st Ionisation Energy : $\mathrm{In} < \mathrm{Al} < \mathrm{Ga} < \mathrm{Tl} < \mathrm{B}$

Choose the correct answer from the options given below:

(A) B and D Only **(B)** A and B Only **(C)** C and D Only **(D)** A and C Only

Q7. MCQ (Single Correct Answer)

Open solution

10 mL of 2 M NaOH solution is added to 20 mL of 1 M HCl solution kept in a beaker. Now, 10 mL of this mixture is poured into a volumetric flask of 100 mL containing 2 moles of HCl and made the volume upto the mark with distilled water. The solution in this flask is :

(A) 0.2 M NaCl solution **(B)** 20 M HCl solution

(C) Neutral solution **(D)** 10 M HCl solution

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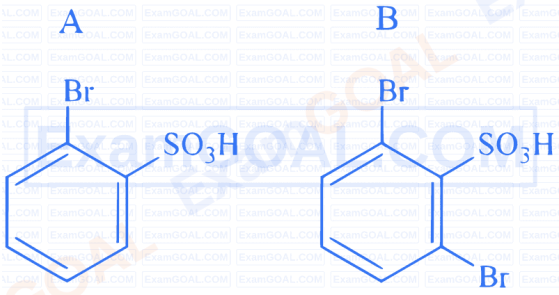
Q8. MCQ (Single Correct Answer)

Open solution

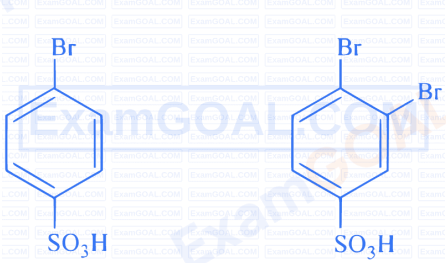
In the following series of reactions identify the major products A & B respectively :



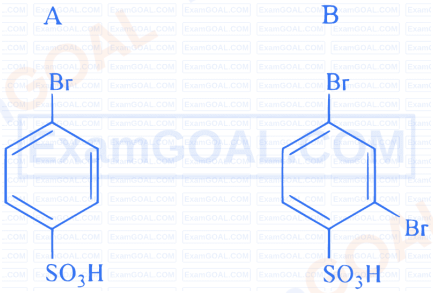
(A)



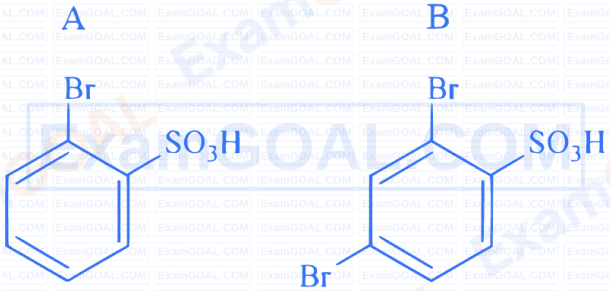
(B)



(C)



(D)



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Q9. MCQ (Single Correct Answer)

Open solution

Identify the diamagnetic octahedral complex ions from below ;

A. $[Mn(CN)_6]^{3-}$

B. $[Co(NH_3)_6]^{3+}$

C. $[Fe(CN)_6]^{4-}$

D. $[Co(H_2O)_3F_3]$

Choose the correct answer from the options given below:

(A) B and C Only **(B)** A and C Only **(C)** B and D Only **(D)** A and D Only

Q10. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I : CrO_3 is a stronger oxidizing agent than MoO_3

Statement II : $Cr(VI)$ is more stable than $Mo(VI)$

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q11. MCQ (Single Correct Answer)

Open solution

Consider the following statements related to temperature dependence of rate constants.

Identify the correct statements.

A. The Arrhenius equation holds true only for an elementary homogenous reaction.

B. The unit of A is same as that of k in Arrhenius equation.

C. At a given temperature, a low activation energy means a fast reaction.

D. A and Ea as used in Arrhenius equation depend on temperature.

E. When $Ea \gg RT$, A and Ea become interdependent.

Choose the correct answer from the options given below:

(A) A, C and D Only **(B)** B, D and E Only **(C)** B and C Only **(D)** A and B Only

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Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Hyperconjugation is not a permanent effect.

Statement II : In general, greater the number of alkyl groups attached to a positively charged Cation, greater is the hyperconjugation interaction and stabilization of the cation.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

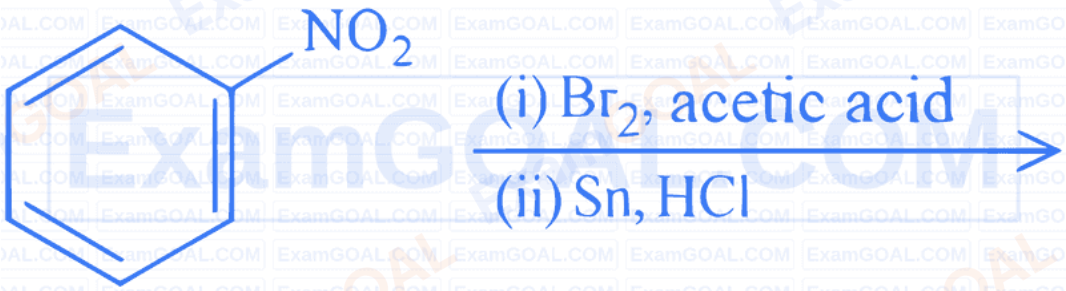
(D) Both Statement I and Statement II are true

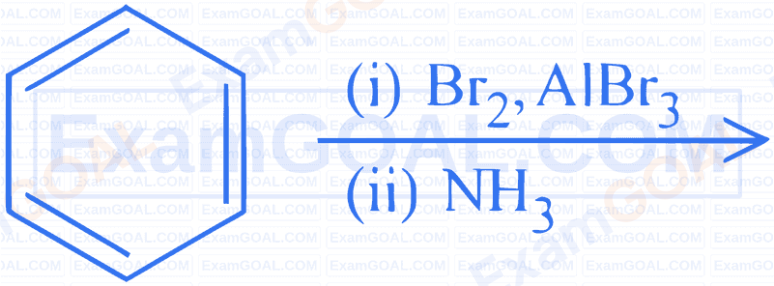
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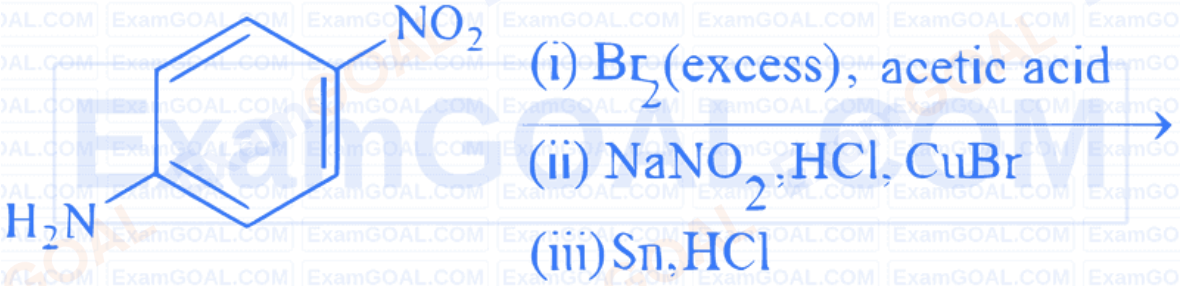
Q13. MCQ (Single Correct Answer)

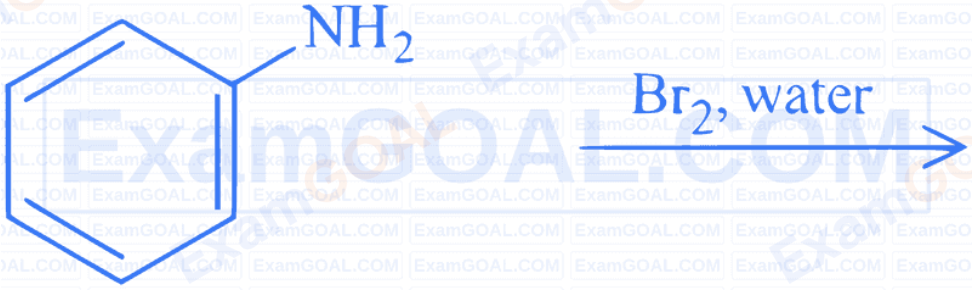
Open solution

The sequence from the following that would result in giving predominantly $3,4,5-$ Tribromoaniline is :

(A) 

(B) 

(C) 

(D) 

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Q14. MCQ (Single Correct Answer)

Open solution

Match the LIST-I with LIST-II

LIST-I (Family)		LIST-II (Symbol of Element)	
A.	Pnictogen (group 15)	I	Ts
B	Chalcogen	II	Og
C	Halogen	III	Lv
D	Noble gas	IV	Mc

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-IV, B-III, C-I, D-II

(C) A-III, B-I, C-IV, D-II

(D) A-II, B-III, C-IV, D-I

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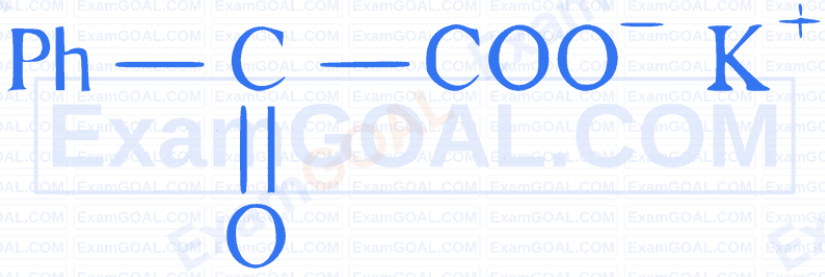
Q15. MCQ (Single Correct Answer)

Open solution

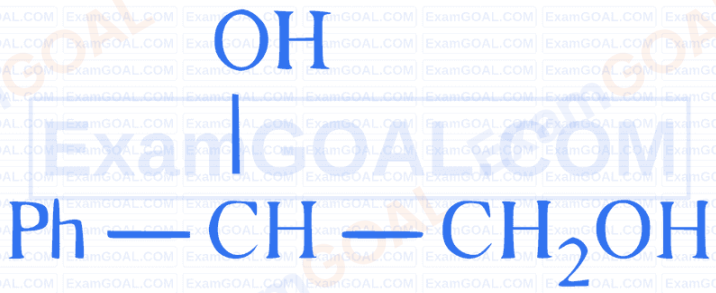
The major product (P) in the following reaction is :



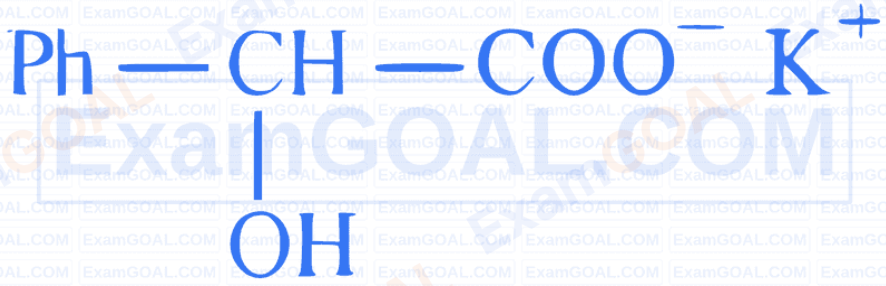
(A)



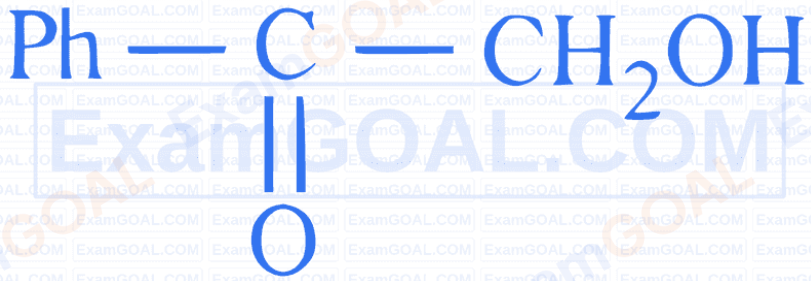
(B)



(C)



(D)



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Q16. MCQ (Single Correct Answer)

Open solution

The standard cell potential $\left(E_{cell}^{\ominus}\right)$ of a fuel cell based on the oxidation of methanol in air that has been used to power television relay station is measured as 1.21 V . The standard half cell reduction potential for O_2 $\left(E_{O_2/H_2O}^{\circ}\right)$ is 1.229 V .

Choose the correct statement :

(A) Reactants are fed at one go to each electrode.

(B) Oxygen is formed at the anode.

(C) The standard half cell reduction potential for the reduction of CO_2 $\left(E_{CO_2/CH_3OH}^{\circ}\right)$ is 19 mV

(D) Reduction of methanol takes place at the cathode.

Q17. MCQ (Single Correct Answer)

Open solution

For electrons in ' 2 s ' and ' 2 p ' orbitals, the orbital angular momentum values, respectively are:

(A) 0 and $\sqrt{6}\frac{h}{2\pi}$

(B) 0 and $\sqrt{2}\frac{h}{2\pi}$

(C) $\sqrt{2}\frac{h}{2\pi}$ and 0

(D) $\frac{h}{2\pi}$ and $\sqrt{2}\frac{h}{2\pi}$

Q18. MCQ (Single Correct Answer)

Open solution

Compounds that should not be used as primary standards in titrimetric analysis are :

A. $Na_2Cr_2O_7$

B. Oxalic acid

C. NaOH

D. $FeSO_4 \cdot 6H_2O$

E. Sodium tetraborate

Choose the most appropriate answer from the options given below:

(A) C, D and E Only

(B) B and D Only

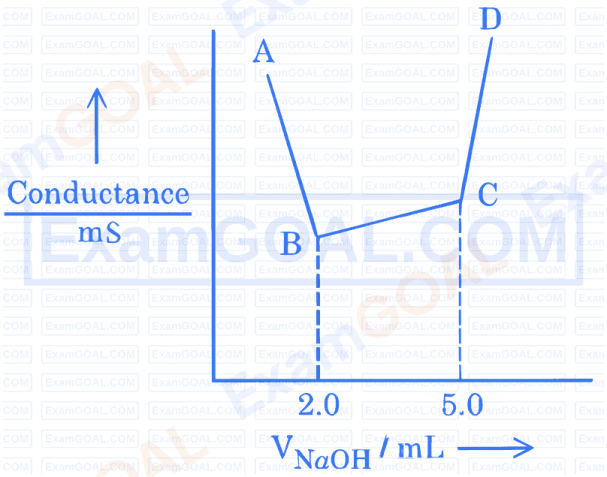
(C) D and E Only

(D) A, C and D Only

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Q19. MCQ (Single Correct Answer)

40 mL of a mixture of CH_3COOH and HCl (aqueous solution) is titrated against 0.1 M NaOH solution conductometrically. Which of the following statement is correct?



Open solution

(A) The concentration of CH_3COOH in the original mixture is 0.005 M

(B) The concentration of HCl in the original mixture is 0.005 M

(C) CH_3COOH is neutralised first followed by neutralisation of HCl

(D) Point ' C ' indicates the complete neutralisation of HCl

Q20. MCQ (Single Correct Answer)

In Dumas' method for estimation of nitrogen 0.4 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is (Given : Aqueous tension at $300\ K = 15\ mmHg$)

(A) 15.71% (B) 17.46% (C) 7.85% (D) 20.95%

Open solution

Q21. Numerical

X g of nitrobenzene on nitration gave 4.2 g of m -dinitrobenzene. X = _____g. (nearest integer)

[Given : molar mass (in $gmol^{-1}$) $C : 12, H : 1, O : 16, N : 14$]

Open solution

Q22. Numerical

The total number of structural isomers possible for the substituted benzene derivatives with the molecular formula C_9H_{12} is_____.

Open solution

Q23. Numerical

Among, Sc, Mn, Co and Cu , identify the element with highest enthalpy of atomisation. The spin only magnetic moment value of that element in its +2 oxidation state is_____BM (in nearest integer).

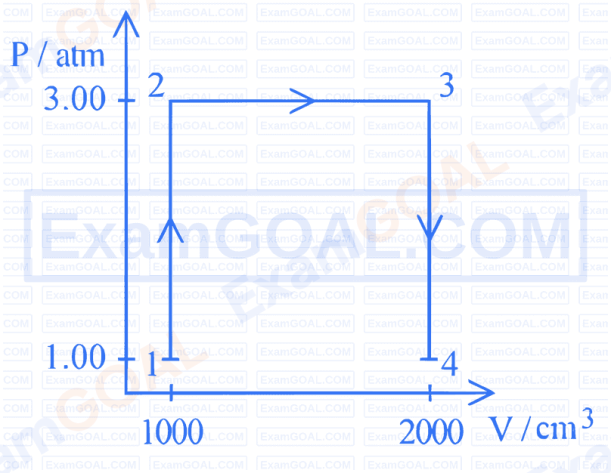
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Q24. Numerical

A perfect gas (0.1 mol) having $\overline{\mathrm{C}}_v = 1.50\mathrm{R}$ (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is $(-)$_____J (nearest integer)

[Given: $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$]



Open solution

Q25. Numerical

A sample of n -octane $(1.14\ \mathrm{g})$ was completely burnt in excess of oxygen in a bomb calorimeter, whose heat capacity is $5\ \mathrm{kJ\ K^{-1}}$. As a result of combustion reaction, the temperature of the calorimeter is increased by 5 K . The magnitude of the heat of combustion of octane at constant volume is_____ $\mathrm{kJmol^{-1}}$ (nearest integer).

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

The area of the region $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$ is

(A) $\frac{512}{3}$ (B) $\frac{2048}{3}$ (C) 512 (D) $\frac{1024}{3}$

Open solution

Q27. MCQ (Single Correct Answer)

Line L_1 of slope 2 and line L_2 of slope $\frac{1}{2}$ intersect at the origin O . In the first quadrant, P_1, $P_2, \ldots, P_{12}$ are 12 points on line L_1 and $Q_1, Q_2, \ldots, Q_9$ are 9 points on line L_2. Then the total number of triangles, that can be formed having vertices at three of the 22 points $\mathrm{O}, \mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_{12}$, $\mathrm{Q}_1, \mathrm{Q}_2, \ldots, \mathrm{Q}_9$, is:

(A) 1026 (B) 1188 (C) 1134 (D) 1080

Open solution

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Q28. MCQ (Single Correct Answer)

Each of the angles β and γ that a given line makes with the positive y - and z-axes, respectively, is half of the angle that this line makes with the positive x-axes. Then the sum of all possible values of the angle β is

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Open solution

Q29. MCQ (Single Correct Answer)

If $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of an equilateral triangle whose centroid is z_0, then $\sum_{k=1}^{3} (z_k - z_0)^2$ is equal to

(A) 0 (B) 1 (C) i (D) -i

Open solution

Q30. MCQ (Single Correct Answer)

Let C be the circle of minimum area enclosing the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with eccentricity $\frac{1}{2}$ and foci $(\pm 2, 0)$. Let PQR be a variable triangle, whose vertex P is on the circle C and the side QR of length $2a$ is parallel to the major axis of E and contains the point of intersection of E with the negative y-axis. Then the maximum area of the triangle PQR is :

(A) $8(3 + \sqrt{2})$ (B) $8(2 + \sqrt{3})$ (C) $6(3 + \sqrt{2})$ (D) $6(2 + \sqrt{3})$

Open solution

Q31. MCQ (Single Correct Answer)

The shortest distance between the curves $y^2 = 8x$ and $x^2 + y^2 + 12y + 35 = 0$ is:

(A) $2\sqrt{3} - 1$ (B) $2\sqrt{2} - 1$ (C) $3\sqrt{2} - 1$ (D) $\sqrt{2}$

Open solution

Q32. MCQ (Single Correct Answer)

Consider the lines $x(3\lambda + 1) + y(7\lambda + 2) = 17\lambda + 5, \lambda$ being a parameter, all passing through a point P. One of these lines (say L) is farthest from the origin. If the distance of L from the point $(3, 6)$ is d, then the value of d^2 is

(A) 10 (B) 20 (C) 15 (D) 30

Open solution

Q33. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation

$\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x, y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{4}{3}$ (B) $\frac{2}{3} + e^3$ (C) $\frac{4}{3} + e^3$ (D) $\frac{2}{3}$

Open solution

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Q34. MCQ (Single Correct Answer)

Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x, x\neq 0$. Then $f(3)+f(8)$ is equal to

(A) 13 (B) 11 (C) 10 (D) 12

Open solution

Q35. MCQ (Single Correct Answer)

Let the equation $x(x+2)(12-k)=2$ have equal roots. Then the distance of the point $\left(k,\frac{k}{2}\right)$ from the line $3x+4y+5=0$ is

(A) 15 (B) 12 (C) $5\sqrt{3}$ (D) $15\sqrt{5}$

Open solution

Q36. MCQ (Single Correct Answer)

Let $A=\{-2,-1,0,1,2,3\}$. Let R be a relation on A defined by $x\mathrm{R}y$ if and only if $y=\max\{x,1\}$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+m+n$ is equal to

(A) 11 (B) 12 (C) 14 (D) 13

Open solution

Q37. MCQ (Single Correct Answer)

Let the Mean and Variance of five observations $x_1=1, x_2=3, x_3=a, x_4=7$ and $x_5=\mathrm{b}, a>\mathrm{b}$, be 5 and 10 respectively. Then the Variance of the observations $n+x_n, n=1,2,\ldots,5$ is

(A) 17 (B) 16 (C) 16.4 (D) 17.4

Open solution

Q38. MCQ (Single Correct Answer)

The integral $\int_0^{\pi}\frac{8xdx}{4\cos^2 x+\sin^2 x}$ is equal to

(A) $2\pi^2$ (B) $4\pi^2$ (C) π^2 (D) $\frac{3\pi^2}{2}$

Open solution

Q39. MCQ (Single Correct Answer)

The distance of the point $(7,10,11)$ from the line $\frac{x-4}{1}=\frac{y-4}{0}=\frac{z-2}{3}$ along the line $\frac{x-9}{2}=\frac{y-13}{3}=\frac{z-17}{6}$ is

(A) 16 (B) 12 (C) 18 (D) 14

Open solution

Q40. MCQ (Single Correct Answer)

If the probability that the random variable X takes the value x is given by

$P(X=x)=k(x+1)3^{-x}, x=0,1,2,3\ldots$, where k is a constant, then $P(X\geq 3)$ is equal to

(A) $\frac{1}{9}$ (B) $\frac{8}{27}$ (C) $\frac{7}{27}$ (D) $\frac{4}{9}$

Open solution

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Q41. MCQ (Single Correct Answer)

The number of solutions of the equation
$(4-\sqrt{3})\sin x - 2\sqrt{3}\cos^2 x = -\frac{4}{1+\sqrt{3}}, x \in \left[-2\pi, \frac{5\pi}{2}\right]$ is

(A) 4 (B) 3 (C) 6 (D) 5

Open solution

Q42. MCQ (Single Correct Answer)

If the domain of the function $f(x) = \log_7\left(1 - \log_4\left(x^2 - 9x + 18\right)\right)$ is $(\alpha, \beta) \cup (\gamma, \delta)$, then $\alpha + \beta + \gamma + \delta$ is equal to

(A) 17 (B) 15 (C) 16 (D) 18

Open solution

Q43. MCQ (Single Correct Answer)

Let $f : \mathrm{R} \to \mathrm{R}$ be a function defined by $f(x) = \big||x+2| - 2|x|\big|$. If m is the number of points of local minima and n is the number of points of local maxima of f, then $m + n$ is

(A) 3 (B) 4 (C) 2 (D) 5

Open solution

Q44. MCQ (Single Correct Answer)

The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \ldots$ upto ∞ terms, is equal to

(A) $3e$ (B) $2e$ (C) $4e$ (D) $6e$

Open solution

Q45. MCQ (Single Correct Answer)

If the four distinct points $(4,6), (-1,5), (0,0)$ and $(k, 3k)$ lie on a circle of radius r, then $10k + r^2$ is equal to

(A) 34 (B) 32 (C) 35 (D) 33

Open solution

Q46. Numerical

If $\lim\limits_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}} = p$, then $96\log_e p$ is equal to _____

Open solution

Q47. Numerical

If the equation of the hyperbola with foci $(4,2)$ and $(8,2)$ is $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, then $\alpha + \beta + \gamma$ is equal to_____.

Open solution

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Q48. Numerical

Let $\left(1+x+x^2\right)^{10}=a_0+a_1x+a_2x^2+\ldots+a_{20}x^{20}$. If $(a_1+a_3+a_5+\ldots+a_{19})-11a_2=121k$, then k is equal to_____ .

Open solution

Q49. Numerical

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}, \vec{b}=3\hat{i}-3\hat{j}+3\hat{k}, \vec{c}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b}\times\vec{d}=\vec{c}\times\vec{d}$ and $\vec{a}\cdot\vec{d}=4$. Then $|(\vec{a}\times\vec{d})|^2$ is equal to_____.

Open solution

Q50. Numerical

Let I be the identity matrix of order 3×3 and for the matrix $A=\begin{bmatrix}\lambda & 2 & 3\\ 4 & 5 & 6\\ 7 & -1 & 2\end{bmatrix}$, $|A|=-1$. Let B be the inverse of the matrix $\operatorname{adj}\left(A\operatorname{adj}\left(A^2\right)\right)$. Then $|(\lambda B+I)|$ is equal to_____

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

Two cylindrical vessels of equal cross sectional area of $2\,\mathrm{m}^2$ contain water upto heights 10 m and 6 m , respectively. If the vessels are connected at their bottom then the work done by the force of gravity is (Density of water is $10^3\,\mathrm{kg/m}^3$ and $\mathrm{g}=10\,\mathrm{m/s}^2$)

(A) $1\times10^5\,\mathrm{J}$ (B) $4\times10^4\,\mathrm{J}$ (C) $8\times10^4\,\mathrm{J}$ (D) $6\times10^4\,\mathrm{J}$

Open solution

Q52. MCQ (Single Correct Answer)

Two monochromatic light beams have intensities in the ratio 1:9. An interference pattern is obtained by these beams. The ratio of the intensities of maximum to minimum is

(A) $8:1$ (B) $4:1$ (C) $3:1$ (D) $9:1$

Open solution

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Q53. MCQ (Single Correct Answer)

Open solution

A particle moves along the x-axis and has its displacement x varying with time t according to the equation:

$x = c_0\left(t^2 - 2\right) + c(t-2)^2$

where c_0 and c are constants of appropriate dimensions.

Then, which of the following statements is correct?

(A) the acceleration of the particle is $2\,(c + c_0)$

(B) the acceleration of the particle is $2c_0$

(C) the acceleration of the particle is 2 c

(D) the initial velocity of the particle is $4c$

Q54. MCQ (Single Correct Answer)

Open solution



Consider two blocks A and B of masses $m_1 = 10\ \text{kg}$ and $m_2 = 5\ \text{kg}$ that are placed on a frictionless table. The block A moves with a constant speed $v = 3\ \text{m/s}$ towards the block B kept at rest. A spring with spring constant $k = 3000\ \text{N/m}$ is attached with the block B as shown in the figure. After the collision, suppose that the blocks A and B, along with the spring in constant compression state, move together, then the compression in the spring is, (Neglect the mass of the spring)

(A) 0.3 m (B) 0.1 m (C) 0.4 m (D) 0.2 m

Q55. MCQ (Single Correct Answer)

Open solution

A magnetic dipole experiences a torque of $80\sqrt{3}\ \text{N m}$ when placed in uniform magnetic field in such a way that dipole moment makes angle of 60° with magnetic field. The potential energy of the dipole is :

(A) $-40\sqrt{3}\ \text{J}$ (B) -80 J

(C) 80 J (D) -60 J

Q56. MCQ (Single Correct Answer)

Open solution

An ideal gas exists in a state with pressure P_0, volume V_0. It is isothermally expanded to 4 times of its initial volume (V_0), then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is

(A) $P_0\,V_0(\ln 2 - 0.75)$ (B) $P_0\,V_0(2\ln 2 - 0.75)$

(C) $P_0\,V_0(2\ln 2 - 0.25)$ (D) $P_0\,V_0(\ln 2 - 0.25)$

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Q57. MCQ (Single Correct Answer)

Width of one of the two slits in a Young's double slit interference experiment is half of the other slit. The ratio of the maximum to the minimum intensity in the interference pattern is :

Open solution

(A) $3:1$

(B) $(2\sqrt{2}+1):(2\sqrt{2}-1)$

(C) $9:1$

(D) $(3+2\sqrt{2}):(3-2\sqrt{2})$

Q58. MCQ (Single Correct Answer)

Using a battery, a 100 pF capacitor is charged to 60 V and then the battery is removed. After that, a second uncharged capacitor is connected to the first capacitor in parallel. If the final voltage across the second capacitor is 20 V , its capacitance is: (in pF)

Open solution

(A) 600

(B) 100

(C) 400

(D) 200

Q59. MCQ (Single Correct Answer)

A motor operating on 100 V draws a current of 1 A . If the efficiency of the motor is 91.6%, then the loss of power in units of $\mathrm{cal/s}$ is

Open solution

(A) 6.2

(B) 2

(C) 8.4

(D) 4

Q60. MCQ (Single Correct Answer)

In the resonance experiment, two air columns (closed at one end) of 100 cm and 120 cm long, give 15 beats per second when each one is sounding in the respective fundamental modes. The velocity of sound in the air column is:

Open solution

(A) $370\ \mathrm{m/s}$

(B) $340\ \mathrm{m/s}$

(C) $335\ \mathrm{m/s}$

(D) $360\ \mathrm{m/s}$

Q61. MCQ (Single Correct Answer)

A block of mass 1 kg , moving along x with speed $v_i = 10\ \mathrm{m/s}$ enters a rough region ranging from $x = 0.1\ \mathrm{m}$ to $x = 1.9\ \mathrm{m}$. The retarding force acting on the block in this range is $\mathrm{F_r} = -\mathrm{kx N}$, with k $= 10\ \mathrm{N/m}$. Then the final speed of the block as it crosses rough region is.

Open solution

(A) $4\ \mathrm{m/s}$

(B) $10\ \mathrm{m/s}$

(C) $8\ \mathrm{m/s}$

(D) $6\ \mathrm{m/s}$

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Q62. MCQ (Single Correct Answer)

Open solution

The truth table corresponding to the circuit given below is:



(A)

A	B	C
0	0	0
0	1	0
1	0	1
1	1	1

(B)

A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

(C)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

(D)

A	B	C
0	0	0
1	0	0
0	1	0
1	1	1

Q63. MCQ (Single Correct Answer)

Open solution

A monochromatic light of frequency 5×10^{14} Hz travelling through air, is incident on a medium of refractive index ' 2 '. Wavelength of the refracted light will be :

(A) 400 nm **(B)** 300 nm **(C)** 600 nm **(D)** 500 nm

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Q64. MCQ (Single Correct Answer)

Open solution

Match the LIST-I with LIST-II

LIST-I		LIST-II	
A.	Boltzmann constant	I	$\mathrm{ML^2\,T^{-1}}$
B	Coefficient of viscosity	II	$\mathrm{MLT^{-3}\,K^{-1}}$
C	Planck's constant	III	$\mathrm{ML^2\,T^{-2}\,K^{-1}}$
D	Thermal conductivity	IV	$\mathrm{ML^{-1}\,T^{-1}}$

Choose the correct answer from the options given below:

(A) A - III, B - IV, C - I, D - II

(B) A - III, B - IV, C - II, D - I

(C) A - III, B - II, C - I, D - IV

(D) A - II, B - III, C - IV, D - I

Q65. MCQ (Single Correct Answer)

Open solution

Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by $1^{\circ}\mathrm{C}$. Its initial temperature must be:

(A) 2500 K

(B) $25^{\circ}\mathrm{C}$

(C) $250^{\circ}\mathrm{C}$

(D) 250 K

Q66. MCQ (Single Correct Answer)

Open solution

A solid steel ball of diameter 3.6 mm acquired terminal velocity $2.45 \times 10^{-2}\ \mathrm{m/s}$ while falling under gravity through an oil of density $925\ \mathrm{kg\,m^{-3}}$. Take density of steel as $7825\ \mathrm{kg\,m^{-3}}$ and g as $9.8\ \mathrm{m/s^2}$. The viscosity of the oil in SI unit is

(A) 2.18

(B) 1.68

(C) 2.38

(D) 1.99

Q67. MCQ (Single Correct Answer)

Open solution

A particle is projected with velocity u so that its horizontal range is three times the maximum height attained by it. The horizontal range of the projectile is given as $\frac{nu^2}{25g}$, where value of n is: (Given, ' g ' is the acceleration due to gravity.)

(A) 6

(B) 12

(C) 18

(D) 24

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Q68. MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as $\mathbf{Assertion A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The Bohr model is applicable to hydrogen and hydrogen-like atoms only.

Reason $\mathbf{R}$: The formulation of Bohr model does not include repulsive force between electrons.

In the light of the above statements, choose the correct answer from the options given below

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Q69. MCQ (Single Correct Answer)

Open solution

An electric bulb rated as $100\ \mathrm{W} - 220\ \mathrm{V}$ is connected to an ac source of rms voltage 220 V. The peak value of current through the bulb is :

(A) 0.32 A (B) 0.64 A (C) 0.45 A (D) 2.2 A

Q70. MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason** $\mathbf{R}$

Assertion A : If Oxygen ion $\left(\mathrm{O}^{-2}\right)$ and Hydrogen ion $\left(\mathrm{H}^{+}\right)$enter normal to the magnetic field with equal momentum, then the path of O^{-2} ion has a smaller curvature than that of H^{+}.

Reason R : A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Q71. Numerical

Open solution

An electron in the hydrogen atom initially in the fourth excited state makes a transition to n^{th} energy state by emitting a photon of energy 2.86 eV . The integer value of n will be_____.

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Q72. Numerical

Light from a point source in air falls on a spherical glass surface (refractive index, $\mu = 1.5$ and radius of curvature $= 50\ \text{cm}$). The image is formed at a distance of 200 cm from the glass surface inside the glass. The magnitude of distance of the light source from the glass surface is _____m.

Open solution

Q73. Numerical

Two cells of emfs 1 V and 2 V and internal resistances 2Ω and 1Ω, respectively, are connected in series with an external resistance of 6Ω. The total current in the circuit is I_1. Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2. The value of $\left(\frac{\mathrm{I}_1}{\mathrm{I}_2}\right)$ is $\frac{x}{3}$. The value of x is_____.

Open solution

Q74. Numerical

The excess pressure inside a soap bubble A in air is half the excess pressure inside another soap bubble B in air. If the volume of the bubble A is n times the volume of the bubble B, then, the value of n is_____.

Open solution

Q75. Numerical

A physical quantity C is related to four other quantities $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s as follows

$C = \frac{pq^2}{r^3\sqrt{s}}$

The percentage errors in the measurement of $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s are $1\%, 2\%, 3\%$ and 2%, respectively. The percentage error in the measurement of C will be_____%

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **A**	4. **D**	5. **C**
6. **A**	7. **B**	8. **B**	9. **A**	10. **A**
11. **C**	12. **A**	13. **C**	14. **B**	15. **C**
16. **C**	17. **B**	18. **D**	19. **B**	20. **A**
21. **3**	22. **8**	23. **4**	24. **304**	25. **2500**

Mathematics

26. **D**	27. **C**	28. **C**	29. **A**	30. **B**
31. **B**	32. **B**	33. **A**	34. **B**	35. **A**
36. **B**	37. **B**	38. **A**	39. **D**	40. **A**
41. **D**	42. **D**	43. **A**	44. **B**	45. **C**
46. **32**	47. **141**	48. **239**	49. **128**	50. **38 or -38**

Physics

51. **C**	52. **B**	53. **A**	54. **B**	55. **B**
56. **B**	57. **D**	58. **D**	59. **B**	60. **D**
61. **C**	62. **A**	63. **B**	64. **A**	65. **D**
66. **D**	67. **D**	68. **B**	69. **B**	70. **C**
71. **2**	72. **4**	73. **4**	74. **8**	75. **15**

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